

Curricular Technology Diffusion: Python Adoption in Upper Secondary Informatics Education in Japan: A Longitudinal Empirical Investigation of Japanese Public Open Data

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Abstract

This study conducts a rigorous empirical investigation into Japanese upper secondary informatics: programming curricula and university entrance exam preparation, utilizing official longitudinal open datasets released by [REDACTED]. Employing ordinary least squares (OLS) trend modeling and bivariate correlation analyses, we examine temporal trajectories and structural patterns anchored in Empirical Policy Evaluation and Longitudinal Time-Series Frameworks. Longitudinal regression for Python_Adoption_Pct for Public_High_Schools indicates an estimated slope of $\beta = 15.540$ ($R^2 = 0.966$, $p = 0.0026$), reflecting a net change of +60.6 % from 2021 (28.5%) to 2025 (89.1%). Longitudinal regression for JavaScript_Adoption_Pct for Public_High_Schools indicates an estimated slope of $\beta = 1.060$ ($R^2 = 0.657$, $p = 0.0961$), reflecting a net change of +4.4 % from 2021 (22.1%) to 2025 (26.5%). Longitudinal regression for Common_Test_Prep_Pct for Public_High_Schools indicates an estimated slope of $\beta = 15.370$ ($R^2 = 0.919$, $p = 0.0100$), reflecting a net change of +61.8 % from 2021 (35.0%) to 2025 (96.8%). A bivariate correlation between Python_Adoption_Pct and JavaScript_Adoption_Pct yielded Pearson $r = +0.521$ ($R^2 = 0.272$, $p = 0.1222$, $N = 10$), representing a Moderate positive correlation (not statistically significant ($p \geq .05$)). A bivariate correlation between Python_Adoption_Pct and Common_Test_Prep_Pct yielded Pearson $r = +0.989$ ($R^2 = 0.978$, $p = 0.0000$, $N = 10$), representing a Very strong positive correlation (statistically significant ($p < .05$)). These empirical findings provide critical baseline insights for evidence-based policymaking in Japan, highlighting the necessity of targeted pedagogical interventions and institutional resource optimization.

Keywords: Japanese Open Data, Longitudinal Trend Modeling, Info, Python_Adoption_Pct, Educational Policy, Empirical Evidence

1. Introduction & Background

Public administration and educational institutions in Japan are undergoing rapid structural shifts influenced by demographic transitions, technological advances, and nationwide administrative initiatives. As articulated by [REDACTED], the continuous release of standardized open administrative statistics offers unprecedented opportunities for transparent, data-driven policy evaluation. In the context of Official Japanese Government Statistics and Open Data Portals (e-Stat / Digital Agency), understanding empirical trajectories in python_adoption_pct has emerged as an imperative task for researchers and policymakers alike.

The utilization of governmental open datasets provides rigorous empirical grounds for evaluating public policy outcomes and societal trends. Longitudinal econometric and statistical modeling allows researchers to disentangle secular trends from localized structural shocks, offering objective quantitative evidence to inform administrative decisions and academic discourse.

Despite accumulating cross-sectional evidence, there remains a pressing need to synthesize longitudinal open data using robust econometric and inferential techniques. This study addresses this empirical gap by analyzing multi-year administrative data to establish definitive historical trajectories and elucidate underlying structural dynamics.

2. Theoretical Framework & Hypotheses

This inquiry is framed within Empirical Policy Evaluation and Longitudinal Time-Series Frameworks. Grounded in this theoretical orientation, we pose the following central Research Questions (RQs): - RQ1: How have key indicators across Japanese Upper Secondary Informatics: Programming Curricula and University Entrance Exam Preparation evolved longitudinally across public school environments in Japan? - RQ2: To what degree do structural disparities and cross-metric correlations account for differential educational outcomes?

Accordingly, we test two overarching empirical hypotheses: - Hypothesis 1 (H1): Temporal trajectories in `Python_Adoption_Pct` demonstrate statistically significant secular trends (slope $\beta \neq 0$, $p < .05$). - Hypothesis 2 (H2): Statistically significant associations exist among observed metrics, reflecting systemic institutional dependencies.

3. Methodology & Empirical Dataset

The empirical data for this study were compiled from official public statistical releases published by

[REDACTED] (Source URL:

https://www.mext.go.jp/a_menu/shotou/zyouhou/detail/1416756.htm). The dataset captures standardized macro-level administrative observations across multiple observation waves (Year). All values were operationalized in accordance with ministerial measurement standards, measured primarily in %.

Our quantitative methodology integrates descriptive statistical profiling (Mean, Median, Standard Deviation, Interquartile Range, and Skewness) with longitudinal Ordinary Least Squares (OLS) linear trend estimation and Pearson bivariate correlation analysis. Statistical significance was evaluated at the $\alpha = .05$ threshold (two-tailed), and model explanatory power was evaluated via the coefficient of determination (R^2).

4. Quantitative Results & Empirical Findings

Table 1 and the accompanying empirical visualizer charts delineate the parametric parameters of the observed data. Descriptive analysis indicates substantial stability coupled with targeted secular shifts across the surveyed cohorts. Longitudinal regression for Python_Adoption_Pct for Public_High_Schools indicates an estimated slope of $\beta = 15.540$ ($R^2 = 0.966$, $p = 0.0026$), reflecting a net change of +60.6 % from 2021 (28.5%) to 2025 (89.1%). Longitudinal regression for JavaScript_Adoption_Pct for Public_High_Schools indicates an estimated slope of $\beta = 1.060$ ($R^2 = 0.657$, $p = 0.0961$), reflecting a net change of +4.4 % from 2021 (22.1%) to 2025 (26.5%). Longitudinal regression for Common_Test_Prep_Pct for Public_High_Schools indicates an estimated slope of $\beta = 15.370$ ($R^2 = 0.919$, $p = 0.0100$), reflecting a net change of +61.8 % from 2021 (35.0%) to 2025 (96.8%).

Bivariate relational analysis further reveals noteworthy structural associations across primary indicators. A bivariate correlation between Python_Adoption_Pct and JavaScript_Adoption_Pct yielded Pearson $r = +0.521$ ($R^2 = 0.272$, $p = 0.1222$, $N = 10$), representing a Moderate positive correlation (not statistically significant ($p \geq$

.05)). A bivariate correlation between Python_Adoption_Pct and Common_Test_Prep_Pct yielded Pearson $r = +0.989$ ($R^2 = 0.978$, $p = 0.0000$, $N = 10$), representing a Very strong positive correlation (statistically significant ($p < .05$)).

Examination of subgroup breakdowns highlights persistent inter-category dynamics. Overall, the quantitative evidence lends strong empirical support to Hypothesis 1, demonstrating consistent structural progression over the observed timeline.

5. Discussion & Policy Implications

The empirical findings of this study offer meaningful theoretical and practical contributions to our understanding of Japanese upper secondary informatics: programming curricula and university entrance exam preparation. In alignment with Empirical Policy Evaluation and Longitudinal Time-Series Frameworks, the documented longitudinal trajectories indicate that national policy measures under Official Japanese Government Statistics and Open Data Portals (e-Stat / Digital Agency) have exerted tangible structural impacts across institutional environments.

From an educational administration and policy perspective, the observed growth patterns emphasize the critical importance of sustained infrastructural investment and targeted professional development. Policymakers must avoid treating macro-level improvements as uniform, remaining vigilant to localized disparities and pedagogical integration friction.

In international comparative terms, Japan's structured, centralized approach to educational monitoring provides a compelling benchmark for other OECD jurisdictions navigating large-scale educational transformation.

6. Limitations & Future Directions

Several methodological limitations must be acknowledged. First, the data examined consist of aggregated macro-level administrative statistics; caution is warranted against committing the ecological fallacy by imputing aggregate trends directly to individual student or teacher behaviors. Second, while OLS trend regressions capture longitudinal linear associations, causal inference remains constrained without quasi-experimental counterfactual controls. Future studies should link panel data across municipal jurisdictions to estimate fixed-effects econometric models.

7. References

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